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Print material container

Abstract

A print material container comprising a body to hold print material, a housing, and a detecting member. The housing being coupled to the body and including a movable mass to move in the housing when a shaking force is applied to the print material container. The detecting member to detect the shaking force and to signal a feedback of the shaking force relative to a shaking force threshold.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2002/0036802	12/2001	Tojo et al.	N/A	N/A
2006/0187743	12/2005	Carreras	366/601	B01F 31/24
2007/0291063	12/2006	Akase et al.	N/A	N/A
2008/0317481	12/2007	Nagano et al.	N/A	N/A
2012/0183311	12/2011	Hasegawa	N/A	N/A
2015/0146496	12/2014	Lucon	366/111	B01J 2/18
2016/0033902	12/2015	Kobayashi	N/A	N/A
2018/0341206	12/2017	Miyanishi et al.	N/A	N/A
2021/0170773	12/2020	Ellis et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1618616	12/2004	CN	N/A
101045403	12/2006	CN	N/A
101329534	12/2007	CN	N/A
101673072	12/2009	CN	N/A
102591174	12/2011	CN	N/A
108859418	12/2017	CN	N/A
108995380	12/2017	CN	N/A
109890622	12/2018	CN	N/A
109939601	12/2018	CN	N/A
4019608	12/1990	DE	N/A
H9-281808	12/1996	JP	N/A
10-2005-0058119	12/2004	KR	N/A

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Background/Summary

BACKGROUND

(1) An image forming apparatus is an apparatus for forming an image on a recording medium or in a space according to an input signal. Representative examples of the apparatus include a printer, a copy machine, a facsimile, a multifunction peripheral (MFP) that integrally implements these functions, a 3D printer, a Biologic printer, an Ink or liquid solution printer, a piezo-electric printer, a pressurized printer, a vacuum printer, a disbursement printer, and derivatives thereof that form images or objects.

(2) One example of an image forming apparatus is a solution that uses an electro-photographic image forming apparatus using an electro-photographic method that forms a visible image utilizing particulates and/or polymers such as toner on a photoconductor by supplying toner to an electrostatic latent image formed on the photoconductor, transfers the toner image by using an intermediate transfer medium, a direct transfer medium, or a chasm-crossing attraction-based transfer medium, and then fixes the transferred toner image on a recording medium.

(3) In other words, the image forming apparatus uses print materials, such as toner, located in a print material container to print an image on the recording medium. As the print material is used in image forming operations, the print material is used up after a certain time period.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram of a configuration of a print material container, according to an example;

(2) FIG. **2** is a view of a housing with a passive mechanical magnetic system to detect shaking, according to an example;

(3) FIG. **3** is a view of a housing with the passive mechanical magnetic system to detect shaking after an shaking force threshold is reached, according to an example;

(4) FIG. **4** is a view of a print material container with a passive mechanical magnetic system to detect shaking, according to an example;

(5) FIGS. **5**A and **5**B are a view of a housing with a passive magnet energy storage to detect shaking, according to an example;

(6) FIG. **6** is a view of a print material container with a passive magnet energy storage to detect shaking, according to an example;

(7) FIG. **7** is a view of the stator coupled to a control board; according to an example;

(8) FIG. **8** is a view of operational logic of the control board, according to an example;

(9) FIGS. **9**A and **9**B are a view of a housing with a detector supply key to feedback to indicate shake detection, according to an example;

(10) FIGS. **10**A and **10**B are a view of the housing with the detector supply key when it is being shook, according to an example;

(11) FIG. **11** is a view of the print material container with the detector supply key before the print material container has been shaken and a view of the print material container after the print material container has been shaken with a shaking force above a shaking force threshold, according to an example;

(12) FIG. **12** is a view of a locking mechanism in the housing with the detector supply key when it is being shook, according to an example;

(13) FIGS. **13**A and **13**B are a view of a housing with an accumulatorless shake detector with a potentiometer to detect shaking, according to an example;

(14) FIG. **14** is a view of a housing with an accumulatorless shake detector with a visual indicator to detect shaking, according to an example;
(15) FIGS. **15**A and **15**B are a view of print material container with a housing that includes a shake detector having liquids to detect shaking, according to an example;
(16) FIG. **16** is a view of an emitter and sensor to detect a color of the liquids in the housing to detect shaking, according to an example;
(17) FIG. **17** is a side view of a print material container to be coupled to an image forming apparatus or intermediary apparatus providing transfer of print material to the image forming apparatus, according to an example;
(18) FIG. **18** is a block diagram of a configuration of an image forming apparatus, according to an example;

DETAILED DESCRIPTION

(19) Various examples of the disclosure will now be described in greater detail with reference to the accompanying drawings, wherein like reference characters denote like elements, Examples to be explained in the following may be modified and implemented in various different forms.

(20) When it is stated in the disclosure that one element is “connected to” or “coupled to” another element, the expression encompasses not only an example of a direct connection or direct coupling, but also a connection with another element interposed therebetween. Further, when it is stated herein that one element “includes” another element, unless otherwise stated explicitly, it means that yet another element may be further included rather than being excluded.

(21) As used herein, including in the claims, the word “or” is used in an inclusive manner. For example, “A or B” means any of the following: “A” alone, “B” alone, or both “A” and “B”.

(22) An “image forming apparatus” refers to a device for printing print data generated from an external source such as from a central or independent computing-function device or mobile device (such as a smartphone) or memory containing devices (such as a “thumb-drive”) via an electrical or data connection on a recoding medium. The recording medium may include, for example, paper such as glossy paper, plain paper, art paper, overhead projector film, cardstock, a personal or industrial application such as onto or within a space or region of space, or onto a host solution delivering chemical processes to support the printed material in biologic application, and the like. Examples of the image forming apparatus described above may include a copier, a printer, a facsimile, a multi-function printer (MFP) of complexly implementing functions thereof through a single device, a **3**D printer, a Biologic printer, an ink or liquid solution printer, a piezo-electric printer, a pressurized printer, a vacuum printer, a disbursement printer, and derivatives thereof that form images or objects. The image forming apparatus may mean all devices capable of performing an image forming task, such as the printer, the scanner, the fax machine, the multi-function printer (MFP), or a display.

(23) A print material container may also be referred to as a print material cartridge.

(24) The print material in a print material container may become compacted during shipping, and may also become highly dispersed in the print material container toward the end of life of the print material container. In order to reduce compaction of the print material before initial use of the print material container in an image forming apparats and/or to consolidate the print material toward the end of life of the print material container, shaking of the print material container may be used. Shaking the print material container prior to inserting the print material container in the printer helps return the print material to its optimal printing condition. However, it is difficult for a user to know how much shaking work gets the print material to its optimal printing condition.

(25) In addition, toward the end of life of the print material container, print material may become unevenly dispersed in the print material container. Shaking the print material container toward the end of life of the print material container may help disperse the print material throughout the print material container which may help most to all of the print material inside the print material container to be used. This helps reduce waste and improves efficiency. However, it is difficult for a

user to know how much shaking work enables the use of most to all of the print material inside the print material container.

(26) It is necessary for the print material container to be shook with enough force to properly reduce compaction of the print material and/or consolidate the print material. The various embodiments discussed below may help determine whether a print material container is shook with a determined force to reduce compaction of the print material and/or consolidate the print material. In addition, the various embodiments discussed below may also indicate whether the print material container is shook with the determined force to reduce compaction of the print material and/or consolidate the print material.

(27) FIG. **1** shows a view of a print material container **10**. The print material container **10** includes a body **12**, housing **20**, and a detecting member **30**. The body **12** is to hold print material **14**, The housing **20** is coupled to the body **12**. The housing **20** includes a movable mass **22** to move in the housing **20** when a shaking force is applied to the print material container **10**. The detecting member **30** is to detect the shaking force and is to signal a feedback of the shaking force relative to a shaking force threshold. The detecting member **30** may be located in the housing **20**, coupled to the housing **20**, and/or located outside of the housing **20**.

(28) It is desirable for the print material container **10** to be shook above a certain threshold prior to inserting the print material container **10** in an image forming apparatus to reduce compaction of the print material **14** before initial use of the print material container **10** in an image forming apparats. The shaking force threshold may be set at a value determined to reduce compaction of the print material **14** to return the print material **14** to a desired printing condition. In addition, is desirable for the print material container **10** to be shook above a certain threshold toward the end of life of the print material container **10** to disperse the print material **14** throughout the print material container **10** which may help most to all of the print material **14** inside the print material container **10** to be used. The shaking force threshold may be set at a value determined to disperse the print material **14** throughout the print material container **10**.

(29) When a shaking force is applied to the print material container **10**, the shaking force causes the movable mass **22** to move in the housing **20**. The detecting member **30** detects the shaking and may also determine whether the shaking force is above the shaking force threshold. The detecting member **30** then signals feedback to indicate whether the applied shaking force is above the shaking force threshold or not.

(30) Since the detecting member **30** signals feedback that indicates whether the applied shaking force is above the shaking force threshold or not, a user will be able to understand if the print material container **10** has been shaken enough or if the print material container **10** needs to be shaken some more and/or with greater force.

(31) FIG. **2** shows an example of the housing **20** coupled to the print material container **10** with a passive mechanical magnetic system to detect shaking as the detecting member **30**.

(32) In this example, the movable mass is a movable magnet **32**. The detecting member **30** includes a first movable conductor **40**, a second movable conductor **41**, and circuitry **42**. The first movable conductor **40**, which is connectable to complete a circuit, is located on an outer edge of the housing **20** on a first side of the movable magnet **32**. The second movable conductor **41**, which is connectable to complete a circuit, is located on the outer edge of the housing **20** on a second side of the movable magnet **32**. The second side is on an opposite of the movable magnet **32** than the first side. The circuitry **42** is located at a first end and at a second end on the outer edge of the housing **20**. The first end and the second end located on opposite sides of the movable magnet **32**.

(33) When the applied shaking force is above the shaking force threshold, the movable magnet **32** is to move the first movable conductor **40** to contact the circuitry **42** at the first end of the housing and to move the second movable conductor **41** to contact the circuitry **42** at the second end of the housing to complete a circuit.

(34) As the print material container **10** is shaken, the movable magnet **32** is forced to move inside

the housing **20**. When the movable magnet **32** comes into contact with either the first movable conductor **40** or the second movable conductor **41**, the moment force from the shaking moves the conductors **40**, **41** a distance by overcoming friction between the movable conductors **40**, **41** and the housing **20**. Once the movable conductors **40**, **41** are moved to a position to contact the circuitry **42**, the circuit is complete. The amount of friction to overcome to move the movable conductors **40**, **21** to contact the circuitry **42** may be set at a threshold that is determined to have a shaking force to move the conductors to be equal to the shaking force needed to properly disperse the print material throughout the print material container **10** either before initial insertion to an image forming apparatus or at the end of life of the print material container **10**.
(35) The print material container **10** may further include a first stationary magnet **46** located at an end of housing **20** on the first side of the movable magnet **32**. The print material container **10** may also further include a second stationary magnet **47** may be located at an end of the housing **20** on the second side of the movable magnet **32**. After the shaking force is depleted, the stationary magnets **46**, **47** may force the movable magnet **32** to the center of the housing **20**.
(36) As an example, the housing **20** may be shaped as a cylinder, a rectangular box, a cube, or a variety of other shapes and sizes. The movable magnet **32** and the stationary magnets **46**, **47** may be shaped to coincide with the shape of the housing **20**.
(37) When the circuit is complete, the detecting member **30** signals the applied shaking force is above the shaking force threshold. The detecting member may send a signal to an image forming apparatus when the print material container **10** is inserted into the image forming apparatus to indicate that the shaking force is above the shaking force threshold. The detecting member **30** may also visually indicate the print material container **10** has been shaken properly. This may be done with different colors, such as green indicating enough shaking force has been applied and red indicating that not enough shaking force has been applied and that the print material container **10** needs to be shook again. However, a variety of different colors may be used to indicate whether the print material container **10** has been shaken enough. Any other visual or audio indicators may also be used to indicate to a user that the print material container **10** has been shake properly. Therefore, a user will be able to understand if the print material container **10** has been shaken enough.
(38) The print material container **10** may further include a removable shipping lock **44** located around the movable magnet **32** to prevent contact between the movable magnet **32** with the first movable conductor **40** and the second movable conductor **41**. The removable shipping lock **44** is to be removed when the print material container **10** is to be inserted into an image forming apparatus and before shaking force is applied to the print material container **10**.
(39) FIG. **3** shows an example of the housing **20** after the movable **32** has moved the first movable conductor **40** to contact the circuitry **42** and has moved the second movable conductor **41** to contact the circuitry **42** to complete the circuit. Since the circuit is complete, the detecting member may then signal the applied shaking force is above the shaking force threshold.
(40) FIG. **4** shows an example of the housing **20** coupled to the print material container **10** with a passive mechanical magnetic system to detect shaking as the detecting member. As can be seen in FIG. **4**, the housing **20** is coupled to the body **12** of the print material container **10**. The print material container **10** may include indicators to shake the print material container in a direction that aligns with the direction the movable magnet **32** is able to move.
(41) FIGS. **5**A and **5**B show an example of the housing **20** coupled to the body of the print material container **10** with a passive magnet energy storage to detect shaking as the detecting member.
(42) In this example, the movable mass is a movable magnet **52**. The detecting member includes a stator **54** located on an outer edge of the housing **20**. When the movable magnet **52** moves in the housing **20** and passes through the stator **54** due to the shaking force, the stator **54** produces an electric current based on the movable magnet **52** moving through the stator **54**. Thus, the stator **54** produces an electric current based on an amount of the shaking force applied to the print material container. The stator **54** may also be field windings.

(43) When the amount of the shaking force is above the shaking force threshold, the stator **54** produces and transmits an electric current above an electric current threshold to indicate the amount of the shaking force is above the shaking force threshold.
(44) The housing **20** may be shaped as a cylinder, a rectangular box, a cube, or a variety of other shapes and sizes. The movable magnet **52** may be shaped to coincide with the shape of the housing **20**.
(45) FIG. **6** shows an example of the housing **20** coupled to the print material container **10** with a passive magnet energy storage to detect shaking as the detecting member. The movable magnet **52** is able to move back and forth though the housing when the print material container **10** is shook. The print material container **10** may include indicators to shake the print material container in a direction that aligns with the direction the movable magnet **52** is able to move.
(46) FIG. **7** shows an example of the housing **20** coupled to the print material container **10** with a passive magnet energy storage to detect shaking as the detecting member. A control board **56** is coupled to the stator to help detect the current produced in the stator **54** when the housing **20** is shook. The control board **56** may include a logic device **57**, a memory **58**, and/or a storage **59**. The storage **59** is to house energy created as the movable magnet **52** moves through the stator **54**. This energy may be used to power the other components of the control board **56**. The logic device **57** may perform computational algorithms to determine if the current is above or below an electric current threshold. The electric current threshold would correspond to the shaking force threshold. The memory **58** may record and save information. The memory **58** may include information detailing a relationship between the electric current threshold and the shaking force threshold. The memory **58** may also store information indicating whether the print material container **10** is shaken above the shaking force threshold. Since the energy used to power the control board **56** is provided by the stator **54** due to the movable magnet **52** moving through the stator **54**, an external power source may not be needed to operate the control board **56** and to detect the shaking of the print material container **10**.
(47) The electric current threshold may be set to correspond to the amount of shaking force to properly disperse the print material throughout the print material container **10** either before initial insertion to an image forming apparatus or at the end of life of the print material container **10**.
(48) FIG. **8** shows an example of operational logic of the control board **56**. When the print material container is installed, the memory **58** is read to determine if an amount of print material in the print material container is below a threshold.
(49) If the amount of print material is above a threshold, the memory **58** is read to determine if the print material container was shaken above the shaking force threshold. If it is determined the print material container was shaken above the shaking force threshold, the control board transmits a signal to indicate the print material container is ready to use. If it is determined the print material container was shaken below the shaking force threshold, the control board transmits a signal to indicate to a user to shake the print material container more.
(50) If the amount of print material is below a threshold that indicates the print material container is at the end of life of the print material container, the memory **58** is read to determine if the print material container was shaken above a shaking force threshold. If it is determined the print material container was shaken above the shaking force threshold, the control board transmits a signal to indicate the print material container is ready to use. If it is determined the print material container was shaken below the shaking force threshold, the control board transmits a signal to indicate to a user to shake the print material container more.
(51) FIGS. **9**A and **9**B show an example of the housing **20** coupled to the body **12** of the print material container **10** with a supply key **94** to indicate whether a shaking force applied to the print material container **10** is above the shaking force threshold. The print material container **10** may include print material **14**.
(52) The detecting member of the print material container **10** includes a supply key **94** located on

the housing **20**. When the applied shaking force to the print material container is above the shaking force threshold; the movable mass **92** contacts and moves the supply key **94** to an end of the housing **20**. When the supply key **94** is moved to the end of the housing **20**, the print material container **10** is insertable into an image forming apparatus.

(53) When the movable mass **92** is shaken; the movable mass contacts the supply key **94**. The force applied to the supply key **94** moves the supply key **94** to the end of the housing. The supply key **94** has resistance against it from moving toward the end of the housing **20**. The resistance may be friction due to the type of material and/or how tightly fit the supply key is located in the housing **20**.

(54) The housing **20** may be shaped as a cylinder, a rectangular box, a cube, or a variety of other shapes and sizes. The movable mass **92** may be shaped to coincide with the shape of the housing **20**. The supply key **94** may be shaped as a cylinder, a rectangular box, a cube, or a variety of other shapes and sizes.

(55) FIGS. **10**A and **10**B show an example of the supply key **94** moving a distance d based on the force the movable mass **92** applies to the supply key **94** and based on the frictional force of the supply **94**. The force the movable mass **92** applies to the supply key **94** is based on the shaking force applied to the print material container **10**.

(56) The amount force to move the supply key **94** to the end of the housing **20** may be set to correspond to the amount of shaking force to properly disperse the print material throughout the print material container **10** either before initial insertion to an image forming apparatus or at the end of life of the print material container **10**.

(57) As can be seen in FIG. **10**, the movable mass **92** moves the supply key **94** toward the end of the housing **20** in the direction d. The print material container **10** may include indicators to shake the print material container in a direction that aligns with the direction the movable mass **92** is able to move.

(58) In FIG. **11**, the print material container **10** is shown with the supply key **94** located on the housing before the print material container has been shaken, and a view of the print material container after the print material container has been shaken with a shaking force above a shaking force threshold.

(59) The location of the supply key on the print material container **10** before the print material container **10** is shook above the shaking force threshold does not enable the print material container **10** to be insertable into the image forming apparatus **96**.

(60) The location of the supply key on the print material container **10** after the print material container **10** is shook above the shaking force threshold enables the print material container **10** to be insertable into the image forming apparatus **96**.

(61) The amount of shaking force to move the supply key **94** to the end of the housing so the print material container **10** is insertable into the image forming apparatus **96** may be set to correspond to the amount of shaking force to properly disperse the print material throughout the print material container **10** either before initial insertion to an image forming apparatus or at the end of life of the print material container **10**.

(62) Therefore, a user may not be able to insert the print material container **10** into the image forming apparatus **96** until the user applies enough shaking force to properly disperse the print material throughout the print material container **10**.

(63) In addition, the location of the supply key **94** may be viewable to a user. The print material container **10** may include indicators that indicate what location the supply key **94** should be moved to. This location would correspond with the amount of shaking force being applied the print material container **10** being above the shaking force threshold. Therefore, when the supply key **94** moves due to the shaking of the print material container **10**, the print material container includes indicators to indicate to a user whether the shaking force applied to the print material container **10** is above the shaking force threshold.

(64) As can be seen in FIG. **12**, the housing **20** may include a locking mechanism **98** to prevent the supply key **94** from returning to its original position when the supply key is moved to the end of the housing **20**. The locking mechanism **98** may be located at the end of the housing **20**. When the print material container **10** reaches an end of life the state, the locking mechanism **98** may be released to enable the supply key **94** to return to its original position. In addition, the release of the lock may also unlock an urging member to enable the urging member to move the supply key **94** back to its original position. This may be caused when the print material level is below a threshold, when the print material container has been used over a certain number of uses, or when the print material container **10** is removed from the image forming apparatus **96**. A variety of other ways may also be used to cause the lock mechanism **98** to disengage.
(65) When the supply key **94** is returned to its original position, then a user may have to apply a shaking force above the shaking force threshold to the print material container **10** to move the supply key **94** back to the end of the housing.
(66) Thus, a user would apply a proper shaking force to the print material container **10** to consolidate the print material toward the end of life of the print material container.
(67) FIGS. **13**A and **13**B show an example of the print material container **10**, where the detecting member **30** includes a slider **134**, a movable mass **132**, and a potentiometer. The slider **134** may be located in the housing **20**. The potentiometer **135** may located on the housing **20** and coupled to the slider **134**. The location of the slider **134** with respect to the potentiometer **135** determines a resistance of the potentiometer.
(68) The housing **20** may further include a spring **133** coupled to the movable mass **132** to cause the movable mass **132** to move back and forth in the housing **20** when the shaking force is applied, wherein when the applied shaking force is above a shaking force threshold. The movable mass **132** is to move the slider **134** to a location in the housing to cause the resistance of the potentiometer **135** to be above a resistance threshold to indicate the shaking force is above the shaking force threshold. The resistance of the potentiometer **135** changes as the slider **134** moves along the potentiometer **135**. When the shaking stops, the spring **133** returns the movable mass **132** to its resting position.
(69) When the print material container **10** is inserted into an image forming apparatus, power is provided and feedback may be given based on the resistance of the potentiometer. The potentiometer **135** may be set so the resistance of the potentiometer **135** increases as the slider **134** is pushed to an end of the housing **20**. The potentiometer **135** may also be set so the resistance of the potentiometer **135** decreases as the slider **134** is pushed to an end of the housing **20**.
(70) The location of the slider **134** and the resistance threshold of the potentiometer **135** may be set to correspond to the amount of shaking force to properly disperse the print material throughout the print material container **10** either before initial insertion to an image forming apparatus or at the end of life of the print material container **10**. The resistance threshold to indicate the shaking force applies is above the shaking force threshold may correspond to the resistance of the potentiometer when the slider **134** is moved to the end of the housing **20**.
(71) When the image forming apparatus receives the feedback from the print material container **10** indicating the resistance of the potentiometer **135**, the image forming apparatus may then indicate whether the print material container **10** needs to be shaken or whether the print material container **10** has been shaken properly.
(72) The housing **20** may be shaped as a cylinder, a rectangular box, a cube, or a variety of other shapes and sizes. The movable mass **132** may be shaped to coincide with the shape of the housing **20**. The slider **134** may be shaped to coincide with the shape of the housing **20**.
(73) The print material container **10** may include indicators to shake the print material container in a direction that aligns with the direction the movable mass **132** is able to move.
(74) The print material container **10** may further include a removable shipping stopper **136** located between the movable mass **132** and the slider **134** in the housing **20** to prevent contact between the

movable mass **132** and the slider **134** before the print material container is to be shook.

(75) FIG. **14** shows an example of a visual indicator **139** coupled to the slider **134**, The visual indicator **139** may indicate the location of the slider to a user without insertion of the print material container into the image forming apparatus. The location of the slider **134** may be set to correspond to the amount of shaking force to properly disperse the print material throughout the print material container **10** either before initial insertion to an image forming apparatus or at the end of life of the print material container **10**. The location of the slider **134** indicating that the amount of shaking force to properly disperse the print material throughout the print material container **10** may be at the end of the housing **20**.

(76) The movable mass **132** may also be coupled to multiple springs and be able to be move along multiple axes. The movable mass **132** may then be able to move multiple sliders, with each slider being coupled to a potentiometer and/or visual indicator.

(77) FIGS. **15**A and **15**B show an example of the print material container **10**, where the detecting member **30** includes a first liquid **153** and a second liquid **154** located in the housing **20**. The second liquid **154** has a different miscibility and/or color than the first liquid **153**. When a shaking force is applied to the print material container **10**, the movable mass **152** agitates the first liquid **153** and the second liquid **154**. When the applied shaking force is above the shaking force threshold, the movable mass **152** agitates the first liquid **153** and the second liquid **154** to cause the first liquid and second liquid to form a mixture **155**. The color and clarity of the mixture is different than the color and clarity of the first liquid **153** and the color and clarity of the second liquid **154**. The color and clarity of the mixture of the first liquid **153** and second liquid **154** may indicate the shaking force is above the shaking force threshold.

(78) The housing may further includes a light port **155** to let in light to reflect off the liquid in the housing so as to be seen by a user through the viewport **156**. The first liquid **153** and the second liquid **154** are not mixed when the print material container **10** is at rest for an extended period of time due to the different miscibility between the first liquid **153** and the second liquid **154**. The amount of shaking force to mix the first liquid **153** and the second liquid **154** to turn to a visible color may be set to correspond to the shaking force threshold, which may correspond to an amount of shaking force to properly disperse the print material throughout the print material container **10** either before initial insertion to an image forming apparatus or at the end of life of the print material container **10**.

(79) After the mixture **155** is formed due to the applied shaking force, the first liquid **153** and the second liquid **154** may reset back to their original form after a period of time has passed. The amount of time for the first liquid **153** and the second liquid **154** reset back to their original form will be dependent on the miscibility of first liquid and the second liquid.

(80) When the applied shaking force is above the shaking force threshold, the first liquid **153** and the second liquid **154** are mixed to turn to a mixture that has a visible color and a high opacity. A user is then able to view this color through the viewport **156** and determine that the print material container **10** has been shaken above the shaking force threshold based on the color of the mixture. The print material container may include indicators indicating what colors indicate the print material container **10** has been shaken enough or indicate the print material container **10** needs to be shaken more.

(81) The combined mixture **155** of the first liquid **153** and the second liquid **154** has distinct optical qualities from the first liquid **153** and the second liquid **154**. Properties of the mixture may be controlled by the choices of the first liquid and the second liquid. In addition properties of the mixture may also be controlled by additive choices, amount of housing space, housing geometry, and free space.

(82) FIG. **16** shows the detecting member may include a light emitter **158** and a sensor **159**. The light emitter **158** is to transmit light to the mixture **155**. The sensor **159** is to detect light reflected from the mixture **155** or light passed through the mixture. When the detected light is a color

corresponding to a color determined to indicate the shaking force applied to the print material container **10** is above the shaking force threshold, the detecting member signals the applied shaking force is above the shaking force threshold.
(83) When the detected light is not a color corresponding to a color determined to indicate the shaking force applied to the print material container **10** is above the shaking force threshold, the detecting member signals the applied shaking force is below shaking force threshold,
(84) FIG. **17** shows an example of a view of a print material container **10** that may be coupled to a first location of a main body of an image forming apparatus **100** to supply print material. As an image forming operation is performed in the image forming apparatus **100**, the print material in the print material container at a location to supply print material is used. The image forming apparatus **100** may include a user interface **200**. The print material container **10** is insertable in the image forming apparatus **100**. In addition, the print material container **10** is installable in the image forming apparatus **100**. The shape of the print material container **10** in FIG. **17** is an example, but the print material container **10** may be various other shapes.
(85) FIG. **18** is a block diagram of a configuration of the image forming apparatus **100**, according to an example.
(86) The image forming apparatus **100** shown in FIG. **18** may include a print material container **10**, the user interface **200**, a memory **220**, and a processor **210**. However, not all the components shown in FIG. **18** are necessary components. The image forming apparatus **100** may be embodied by using less or more components than the components shown in the drawings. Hereinafter, the components will be described.
(87) The processor **210** may control all operations of the image forming apparatus **100** and include at least one processor like a central processing unit (CPU). The processor **210** may control other components in the image forming apparatus **100** to perform operations corresponding to the user input received through the user interface **200**. The processor **210** may include at least one specialized processor corresponding to functions or an all-in-one type processor.
(88) The processor **210** may perform the operational logic shown in FIG. **8**. The processor **210** may also receive the feedback signaled from the detecting member and perform operational logic based on the received to signal to determine whether the print material container **10** has been shaken enough. The user interface **200** may indicate to a user whether the print material container **100** has been shaken enough. In addition, the image forming apparatus **100** may send a signal to user equipment to notify whether the print material container **10** has been shaken enough.
(89) The operating method of the image forming apparatus **100** may be embodied in the form of instructions stored on a machine-readable medium and executable by a computer or a processor. The method of operating the image forming apparatus **100** may be written as computer programs and may be implemented in general-use digital computers that execute the programs using a machine readable recording medium. The above-mentioned machine readable recording medium may be read-only memory (ROM), random-access memory (RAM), flash memory, CD-ROMs, CD-Rs, CD+Rs, CD-RWs, CD+RWs, DVD-ROMs, DVD-Rs, DVD+Rs, DVD-RWs, DVD+RWs, DVD-RAMs, BD-ROMs, BD-Rs, BD-R LTHs, BD-REs, a magnetic tape, floppy disc, a magnet optical recording medium, an optical data recording medium, hard disc, solid-state disc (SSD), and any kind of device capable of storing instructions of machine readable instructions, relevant data, data files, and data structure and capable of providing instructions or machine readable instructions, relevant data, data files, and data structures to a processor and a computer such that the processor or computer may execute the instruction.
(90) The foregoing examples are merely examples and are not to be construed as limiting the disclosure. The disclosure can be readily applied to other types of apparatuses. Also, the description of the examples of the disclosure is intended to be illustrative, and not to limit the scope of the claims.
(91) While the disclosure has been described with reference to the accompanying drawings, it is to

be understood that the scope of the disclosure is defined by the claims described hereinafter and should not be construed as being limited to the above-described examples and/or drawings. It is to be clearly understood that improvements, changes, and modifications that are obvious to those skilled in the art are also within the scope of the disclosure as defined in the claims.

Claims

1. A print material container comprising: a body to hold print material; a housing coupled to the body, the housing including a movable mass to move in the housing when a shaking force is applied to the print material container; and a detecting member to detect the shaking force and to signal a feedback of the shaking force relative to a shaking force threshold, wherein the detecting member includes: a supply key located on the housing, wherein, when the applied shaking force is above the shaking force threshold, the movable mass moves the supply key to an end of the housing.
2. An image forming apparatus, comprising: a device to produce images or objects on a recording medium; and a print material container coupled to the device, wherein the print material container comprises: a body to hold print material; and a housing coupled to the body, the housing including a movable mass to move in the housing when as shaking force is applied to the print material container; and a detecting member to detect the shaking force and to signal a feedback of the shaking force relative to a shaking force threshold, wherein the detecting member includes: a supply key located on the housing, wherein, when the applied shaking force is above the shaking force threshold, the movable mass moves the supply key to an end of the housing.
3. The print material container of claim 1, wherein when the supply key is moved to the end of the housing, the print material container is insertable into an image forming apparatus.
4. The print material container of claim 1, further comprising a locking mechanism located at the end of the housing to prevent the supply key from returning to its original position when the supply key is moved to the end of the housing.
5. The print material container of claim 4, wherein the locking mechanism is releasable when the print material container reaches an end-of-life state.
6. The print material container of claim 1, wherein the supply key is visible to a user, and the print material container includes indicators that indicate a proper location for the supply key corresponding to an applied shaking force above the shaking force threshold.
7. The print material container of claim 1, wherein the housing is cylindrical in shape, and the movable mass is shaped to coincide with the cylindrical shape of the housing.
8. The print material container of claim 1, wherein the print material container includes indicators to instruct a user to shake the print material container in a direction that aligns with a direction in which the movable mass is able to move.
9. The print material container of claim 1, wherein the shaking force threshold corresponds to an amount of shaking force required to properly disperse the print material throughout the print material container.
10. The print material container of claim 1, wherein the movable mass is shaped to correspond with an interior shape of the housing, and the supply key is positioned to receive direct contact from the movable mass when the shaking force is applied.
11. The print material container of claim 1, wherein the housing includes a viewport, and the location of the supply key is visible through the viewport to indicate to a user whether the applied shaking force is above the shaking force threshold.
12. The image forming apparatus of claim 2, further comprising a user interface to indicate to a user whether the print material container has been shaken with a force above the shaking force threshold based on the feedback signaled from the detecting member.
13. The image forming apparatus of claim 2, further comprising a processor to receive the feedback

signaled from the detecting member and determine whether the print material container has been shaken adequately based on the position of the supply key.

14. The image forming apparatus of claim 2, wherein the image forming apparatus is configured to prevent operation when the supply key is not positioned at the end of the housing, indicating the print material container has not been shaken with a force above the shaking force threshold.

15. The image forming apparatus of claim 2, wherein the print material container further comprises a locking mechanism configured to lock the supply key at the end of the housing when the shaking force above the shaking force threshold is applied, and to release the supply key when the print material in the print material container falls below a predetermined threshold level.

16. The image forming apparatus of claim 2, wherein the image forming apparatus is configured to prompt a user to shake the print material container when the feedback from the detecting member indicates the shaking force is below the shaking force threshold.

17. The image forming apparatus of claim 2, wherein the detecting member is configured to signal the feedback to the image forming apparatus when the print material container is inserted into the image forming apparatus.

18. The image forming apparatus of claim 2, wherein the supply key is positioned such that it physically prevents the print material container from being properly inserted into the image forming apparatus until the shaking force above the shaking force threshold has been applied.

19. The image forming apparatus of claim 2, wherein the image forming apparatus is configured to transmit a signal to user equipment to notify whether the print material container has been shaken with a force above the shaking force threshold.